

■ RELEASE ENGINEERING / EXECUTABLE EVIDENCE

A green build. A broken handover.

Old code and new code share one database.
Cutover rehearses the writes between migration steps.

THE MISSED WINDOW

- 01 **ADD COLUMN**
Both versions can still write.
- 02 **BACKFILL**
Existing rows are copied.
- ! **OLD WORKER WRITES**
New column stays stale.
- 03 **ADD SYNC**
Protection arrives too late.

76/76

completed-rollout probes pass

16

migration-window probes fail

v0.3 / SQLITE

01 / PROBLEM

The release is not one moment.

A new-version test can pass while an old worker is still writing.

GitLab documented canary QA passing while older production instances failed inserts.

Source: docs.gitlab.com/development/multi_version_compatibility/

01

ADD COLUMN

The new field now exists.

02

BACKFILL

Rows are copied once.

03

INSTALL SYNC

The two fields now agree.

THE GAP: old worker writes after backfill

The new field is stale even though the migration succeeds.

02 / REPRODUCED COUNTEREXAMPLE

The plan looked green. The data disagreed.

76/76

completed-rollout probes passed

32/48

migration-window probes passed

One failing replay, simplified

ADD COLUMN

BACKFILL

OLD WRITE

ADD SYNC

Expected latest address: 18 Marina Road

Observed by new reader: 4 Broad Street

Make the handover executable.

FIXED OLD CONTRACT

Old reads, updates and inserts

CANDIDATE PLAN

Migration + new SQL adapter

INDEPENDENT LEDGER

Every acknowledged write

Fresh SQLite database for each schedule and string input.

Old writes are inserted at every migration statement boundary.

OUTPUT

Exact SQL + inputs + expected/observed values + replay prefix.

Passing means only these contracts and schedules passed.

What changed the verdict?

Candidate	Completed	Windows
Direct rename	16/76	8/16
One-time backfill	28/76	16/24
Late bridge	76/76	32/48
Window-safe bridge (reference)	76/76	48/48

Curated unsafe patterns caught: baseline 0/3 | rollout 2/3 | windows 3/3

Pre-event reference; curated samples, not Bob output or a customer benchmark.

05 / VALIDATED CLI IMPORT

Bring your own release contract.

WAREHOUSE / BIN RELOCATION

Old scanner workers update pick_bin while new workers read fulfillment_bin.

Late bridge **108/124** **BLOCKED**

Window-safe ordering **124/124** **PASS**

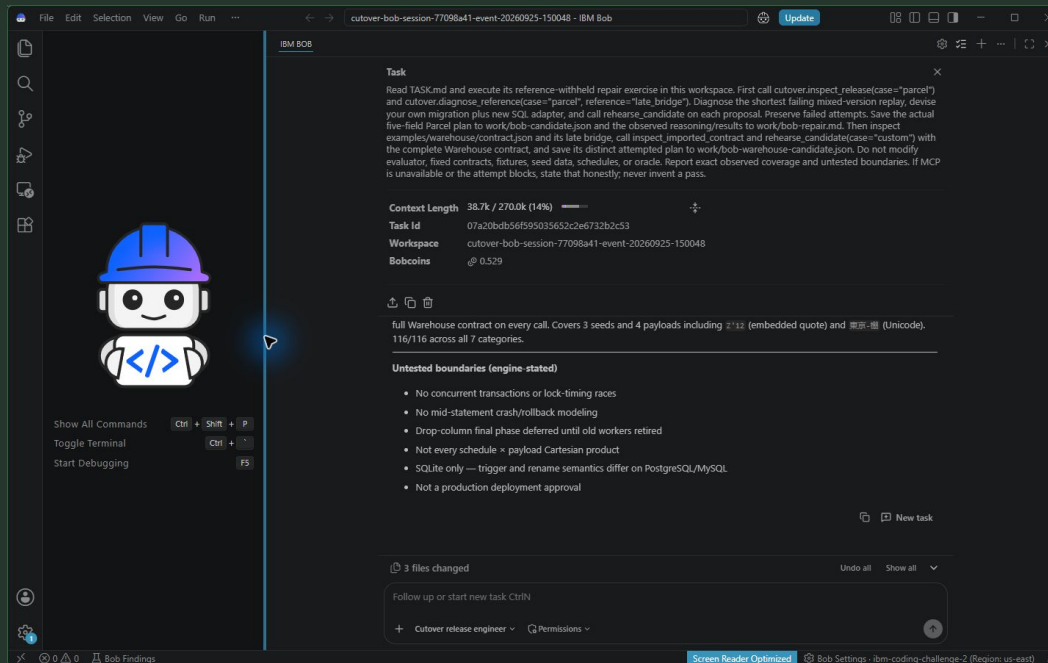
Old write acknowledged: R-07 / New read observed: A-01

JSON + Markdown carry the same executed report, SQL, inputs and hashes.

Pre-event Codex importer and reference plans; Bob repair is shown on the next slide.

Bob repaired the missed window.

ACTUAL BOB TASK SUMMARY



Task: 07a20bdb56f595035652c2e6732b2c53

Candidate: Triggers before backfill,...

Verdict: 116/116 probes

Windows: 40/40

BOB CONTRIBUTION

Found the missed old write; kept a failed trigger attempt; saved a one-way bridge with dual writes.

Plan SHA: 4d54ca78ffc24cc...

A review step for schema-changing PRs.

USER

Backend / release engineer

Shipping a schema change with old workers alive.

VERIFIED WORKFLOW

PR review + CI rehearsal

Attach a witness and bounded evidence.

Revenue hypothesis: paid per-repository CI checks; demand unvalidated.

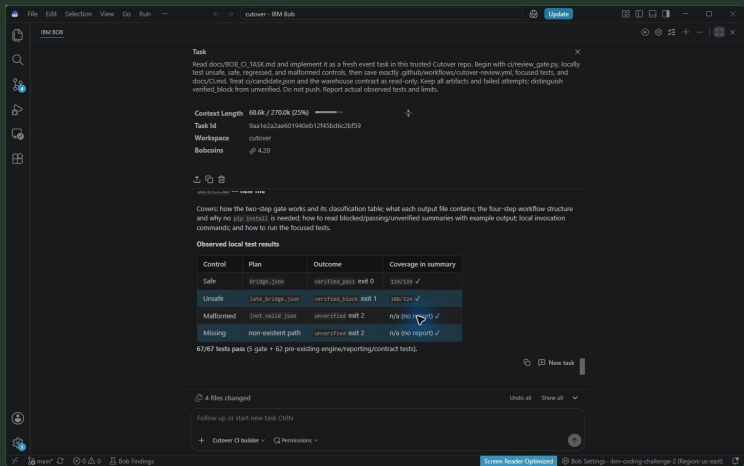
Next: release-team interviews and a real repository adapter.

Limits: SQLite single-table contracts; no concurrency, locks, mid-statement failure, performance or production safety claim.

github.com/josepha-mayo/cutover

A PR gate that keeps the counterexample.

ACTUAL BOB IDE TASK SUMMARY



UNSAFE / RED

108/124 probes | run #36165105646

SAFE REFERENCE / GREEN

124/124 probes | run #36165215088

REGRESSION / RED

100/116 probes | run #36165290677

Task 9aa1e2a2ae601940eb12f45bd6c2bf59

Every run retained JSON, Markdown, ZIP and an independently replayed report.

Controls use pre-event synthetic plans; the PR gate was built in Bob.